Supplementary Information

I. Additional Dataset Statistics

This section provides additional distribution statistics for the two soft-gripper configurations used in Soma. Both the two-finger (2F) and three-finger (3F) configurations cover all 100 object instances. Table S1 reports the corresponding sample counts, proportions, and object-level distribution evenness.

The two configurations contribute nearly equal numbers of samples, and both exhibit high object-level evenness. These statistics indicate comparable object coverage across the two soft-gripper configurations.

II. Parameter Effect and Diagnostic Analysis

This section provides additional diagnostic analyses of the parameter settings used in SomaAlign. These experiments examine the influence of modality-specific masking, conflict-aggregation parameters, and geometry-preservation strength. Unless otherwise stated, the remaining experimental settings are identical to those used in the main paper.

A. Effect of Modality-Specific Masking

To assess the influence of masking probabilities on reconstruction-derived reliability, we retrain Stage I using a uniform probability of 0.583 for all six modalities, approximately the unweighted mean of the default probabilities. Table S2 compares the resulting estimates with those obtained under modality-specific masking.

IMU and tactile retain relatively high reliability under both settings, while Remote RGB and Text remain at the lower end. However, the absolute values change, and Depth and Hand RGB exchange their relative ranking. Thus, predefined masking probabilities influence the estimates but do not solely account for the observed cross-modal differences.

B. Effect of Conflict-Aggregation Parameters

We examine the conflict-aggregation parameters ρ and K_{t} using controlled eight-epoch Stage-III runs with the same Stage-II checkpoint, ECDF calibration, reference centers, random seed, and fixed training and validation subsets. Conflict statistics are evaluated on the same ordered validation subset.

As shown in Table S3, increasing ρ from 0 to 1 raises the mean conflict estimate from 0.3256 to 0.7728 and reduces the refinement coefficient, consistent with greater emphasis on highly conflicting modality pairs. Increasing K_{t} from 1 to 5 includes less extreme pairs in the Top-K average, reducing mean conflict from 0.5940 to 0.5291 and increasing refinement strength.

C. Effect of Geometry-Preservation Strength

We vary the center-protection weight β while keeping $\lambda_{\mathrm{tol}} = 0.1$. We measure normalized center drift, $\|\bar{\mathbf{c}}_i - \mathbf{c}_i\|_2 / D_{\mathrm{global}}$, and the proportion of samples exceeding the prescribed tolerance, which activates the center-protection penalty.

Increasing β from 0.10 to 0.75 reduces mean normalized drift from 0.1018 to 0.0841 and the fraction of samples exceeding the tolerance from 38.8% to 23.9% (Table S4). These results are consistent with stronger reference-center anchoring at larger β.

TABLE S1
SAMPLE DISTRIBUTION ACROSS THE TWO SOFT-GRIPPER CONFIGURATIONS. EVENNESS IS COMPUTED OVER THE 100 OBJECT INSTANCES WITHIN EACH SUBSET.

Gripper	**Objects**	**Samples**	**Proportion**	**Evenness**
2F	100	37,492	50.05%	0.9926
3F	100	37,422	49.95%	0.9973

TABLE S2
MODALITY RELIABILITY UNDER UNIFORM AND MODALITY-SPECIFIC MASKING.

Modality	Uniform 0.583	Modality-specific
Remote RGB	0.095	0.097
Hand RGB	0.399	0.350
Depth	0.518	0.144
Tactile	0.702	0.655
IMU	0.945	0.804
Text	0.020	0.039

TABLE S3
EFFECTS OF CONFLICT-AGGREGATION PARAMETERS ON VALIDATION CONFLICT STATISTICS AND REFINEMENT STRENGTH. P5/P95 DENOTE THE 5TH AND 95TH PERCENTILES OF γ_i. BOLD INDICATES THE DEFAULT SETTING.

Parameter	Setting	γ_i Mean±Std	P5/P95	$1-\bar{\gamma}_B$
ρ	0	0.3256±0.1454	0.1052/0.5743	0.6744
	0.25	0.4403±0.1588	0.1499/0.6686	0.5597
	0.5	0.5630±0.1666	0.2064/0.7517	0.4370
	0.75	0.6773±0.1652	0.2969/0.8182	0.3227
	1.0	0.7728±0.1569	0.3897/0.8737	0.2272
K_{t}	1	0.5940±0.1416	0.2789/0.7534	0.4060
	3	0.5630±0.1666	0.2064/0.7517	0.4370
	5	0.5291±0.1834	0.1696/0.7461	0.4709

TABLE S4
EFFECT OF β ON CENTER DRIFT NORMALIZED BY D_{global}. ACTIVATION DENOTES THE PERCENTAGE OF SAMPLES EXCEEDING $\lambda_{\mathrm{tol}} = 0.1$. BOLD INDICATES THE DEFAULT SETTING.

β	Drift Mean	Drift P95	Activation (%)	$\beta\mathcal{L}_{\mathrm{center}}$
0.10	0.1018	0.1790	38.8	0.0480
0.25	0.0937	0.1697	32.0	0.0919
0.50	0.0877	0.1636	26.6	0.1552
0.75	0.0841	0.1595	23.9	0.2039

Across the tested parameter ranges, the corresponding short-horizon retrieval performance varies by less than 0.4 percentage points within each sweep. Thus, these diagnostic runs show changes in the intended internal quantities alongside comparatively small changes in retrieval performance.

D. Effects of Center-Drift Tolerance and Conflict-Consistency Weight

The center-drift tolerance λ_{tol} determines when the center-protection penalty becomes active, whereas the conflict-consistency weight λ_{conf} scales the auxiliary supervision of the conflict scorer. We examine their effects through controlled eight-epoch Stage-III runs, varying one parameter at a time while keeping the Stage-II initialization, training and validation subsets, random seed, and calibration references fixed.

For $\lambda_{\mathrm{tol}} \in \{0.05, 0.10, 0.20\}$, increasing the tolerance reduces the penalty activation rate from 95.70% to 3.90% and increases mean normalized center drift from 0.0919 to 0.1119, consistent with a wider tolerance allowing greater displacement before the penalty is activated. The default tolerance of 0.10 yields an activation rate of 32.00%. The corresponding R@1 and R@10 spans are 0.20 and 0.32 percentage points, respectively.

For $\lambda_{\mathrm{conf}} \in \{0, 0.05, 0.10, 0.20\}$, conflict-prediction MAE remains between 0.0645 and 0.0647, while R@1 and R@10 span 0.24 and 0.14 percentage points, respectively. The average conflict score and refinement coefficient also change little, indicating limited sensitivity to this auxiliary-loss weight over the tested training horizon. Setting $\lambda_{\mathrm{conf}} = 0$ removes only the conflict-consistency loss; conflict scoring and batch-level modulation remain active.

Together, these results show that the center-drift tolerance controls the activation of geometric constraints as intended, while retrieval performance changes little across the tested parameter settings during the eight-epoch refinement.

III. ADDITIONAL ROBUSTNESS RESULTS

This section provides the complete cross-modal retrieval results under tactile and IMU corruption. Gaussian noise is added to one sensor modality at a time, with ν controlling the noise magnitude relative to the standard deviation of the corresponding clean input tensor. We evaluate each corruption condition using five noise seeds, while keeping the trained checkpoints, the 500 test samples, and the retrieval protocol fixed. Identical perturbations are used across methods for each seed. Recall is macro-averaged over all 30 directed retrieval tasks. Results at $\nu > 0$ are reported as mean $\pm$ sample standard deviation across the five noise seeds, while $\nu = 0$ denotes the fixed clean reference. Performance retention is defined as corrupted Recall divided by the corresponding clean Recall at $\nu = 0$.

As shown in Table S5, SomaAlign maintains higher mean R@1 and R@10 than Pairwise InfoNCE and Hub Alignment throughout the tested corruption sweep. The sample standard deviations do not exceed 0.27 percentage points across all methods and corruption conditions, indicating limited variation across noise realizations under the fixed-checkpoint evaluation. Tactile perturbation produces a relatively sharp initial decline followed by a plateau, whereas IMU perturbation leads to a more gradual degradation as ν increases. At the largest corruption level, SomaAlign also retains a larger fraction of its clean retrieval performance than both controlled baselines. The standard deviations of SomaAlign's R@1/R@10 retention at $\nu = 0.70$ are 0.11/0.10 percentage points under tactile corruption and 0.08/0.17 percentage points under IMU corruption.

TABLE S5
CROSS-MODAL RETRIEVAL UNDER TACTILE AND IMU CORRUPTION. VALUES AT $\nu > 0$ ARE MEAN $\pm$ SAMPLE STANDARD DEVIATION OVER FIVE NOISE SEEDS, EXPRESSED IN PERCENT. RESULTS AT $\nu = 0$ ARE FIXED CLEAN REFERENCES.

Sensor	ν	Pairwise InfoNCE		Hub Alignment		SomaAlign	
		R@1	R@10	R@1	R@10	R@1	R@10
Tactile	0.00	42.00	65.77	44.55	70.55	63.04	90.66
	0.01	32.19 ± 0.04	52.38 ± 0.06	34.55 ± 0.05	56.73 ± 0.07	50.80 ± 0.09	74.48 ± 0.13
	0.03	30.00 ± 0.04	47.67 ± 0.17	32.13 ± 0.05	51.67 ± 0.18	47.77 ± 0.05	68.42 ± 0.20
	0.05	29.82 ± 0.07	47.30 ± 0.13	31.94 ± 0.08	51.30 ± 0.15	47.69 ± 0.07	67.89 ± 0.13
	0.10	29.58 ± 0.05	46.24 ± 0.07	31.75 ± 0.05	50.22 ± 0.13	47.42 ± 0.07	66.60 ± 0.13
	0.30	29.47 ± 0.02	46.02 ± 0.08	31.64 ± 0.03	49.79 ± 0.11	47.27 ± 0.03	66.30 ± 0.08
	0.70	29.53 ± 0.08	46.07 ± 0.04	31.65 ± 0.06	49.91 ± 0.04	47.32 ± 0.07	66.50 ± 0.09
IMU	0.00	42.00	65.77	44.55	70.55	63.04	90.66
	0.01	40.25 ± 0.13	63.29 ± 0.11	42.47 ± 0.11	67.75 ± 0.06	60.43 ± 0.10	87.06 ± 0.05
	0.03	37.71 ± 0.08	58.99 ± 0.17	40.03 ± 0.09	63.60 ± 0.06	57.22 ± 0.08	81.51 ± 0.09
	0.05	36.36 ± 0.17	55.72 ± 0.15	38.55 ± 0.10	60.45 ± 0.24	55.59 ± 0.24	77.91 ± 0.27
	0.10	33.38 ± 0.04	50.44 ± 0.23	35.35 ± 0.10	55.17 ± 0.13	52.16 ± 0.14	72.72 ± 0.24
	0.30	31.44 ± 0.06	46.44 ± 0.18	33.17 ± 0.13	50.40 ± 0.25	49.69 ± 0.06	67.31 ± 0.19
	0.70	31.00 ± 0.04	45.02 ± 0.15	32.63 ± 0.07	48.76 ± 0.20	49.21 ± 0.05	66.13 ± 0.16

TABLE S6
DETAILED PROBE CLASSIFICATION PERFORMANCE ON FIVE DOWNSTREAM TASKS (%). NUMBERS IN PARENTHESES INDICATE CLASS COUNTS. MLP RESULTS ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION OVER FIVE PROBE SEEDS.

Attribute	Linear		MLP	
	Acc	F1	Acc	F1
Hardness (5)	94.91	94.08	96.68±0.27	96.16±0.36
State (2)	91.91	89.07	92.90±0.21	90.40±0.33
Material (10)	94.59	94.87	96.34±0.54	96.02±0.73
Size (6)	94.15	94.19	96.29±0.20	95.51±0.59
Shape (14)	93.64	90.88	95.07±0.50	92.07±1.06
Mean	**93.84**	**92.62**	**95.45**	**94.03**

IV. ADDITIONAL DOWNSTREAM CLASSIFICATION RESULTS

This section provides the detailed task-level results for the frozen-feature downstream classification experiment reported in the main paper. Frozen 1024-dimensional non-text features extracted by SomaAlign are evaluated using logistic regression and a two-layer MLP with a 512–256 architecture and ReLU activations. Independent classifiers are trained for hardness, grasping state, material, size, and shape. Text features are excluded from the probe inputs.

As shown in Table S6, both probe architectures achieve strong performance across all five tasks. The linear probe reaches a mean accuracy and F1-score of 93.84% and 92.62%, respectively. The MLP improves these values to 95.45% and 94.03%. The MLP improves accuracy on all five tasks, while the task-level results also reveal differences in classification difficulty across semantic attributes.

V. ADDITIONAL SOMA-SUCCESS RESULTS

This section provides confusion matrices for the frozen-encoder grasp-outcome evaluation on Soma-Success. Linear and two-layer MLP probes are evaluated using five-fold cross-validation grouped by object identity, with all trials of each object assigned to the same fold. For each fold, the probes are trained on the other four folds and evaluated on the held-out objects.

The confusion matrices in Fig. S1 pool the out-of-fold predictions across all five folds. Each of the 998 valid trials contributes exactly one prediction from a probe trained without any trials of the corresponding object. Each matrix therefore contains 398 failed and 600 successful grasps.

VI. ADDITIONAL CROSS-MODAL GENERATION RESULTS

This section provides qualitative results for the leave-one-out cross-modal generation analysis reported in the main paper. For each target modality, aligned features from the remaining five modalities are used as conditioning inputs to reconstruct the missing sensory stream. Hand RGB, Remote RGB, Depth, and Tactile outputs can be directly visualized as images or pressure maps, whereas IMU is evaluated using the same conditional-generation setting but is not shown here because its temporal outputs are less intuitive to visualize in the same format.

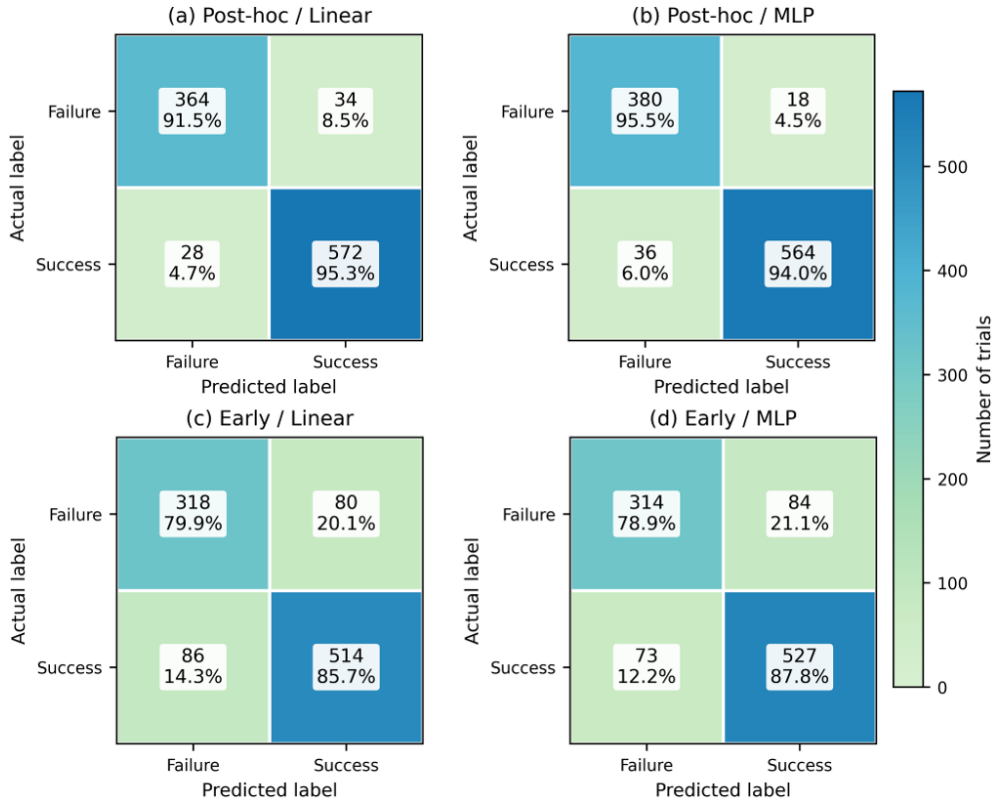


Fig. S1. Confusion matrices for SomaAlign probes on Soma-Success, obtained by pooling out-of-fold predictions from five-fold cross-validation grouped by object identity. Each matrix includes all 998 valid trials, with each trial evaluated once on its held-out fold. Cells report sample counts and row-normalized percentages. (a) Post-hoc linear probe, (b) post-hoc MLP probe, (c) early-prediction linear probe, and (d) early-prediction MLP probe.

As shown in Fig. S2, the generated observations preserve the main object geometry and tactile contact patterns, while some fine-grained visual and pressure details are smoothed. These qualitative examples complement the quantitative semantic-retention analysis reported in the main paper.

VII. Additional Multimodal Question-Answering Results

This section provides additional quantitative and qualitative results for the multimodal question-answering experiment reported in the main paper. The LLaMA-based language model is conditioned on SomaAlign features extracted from Remote RGB, Hand RGB, Depth, Tactile, and IMU observations together with a natural-language prompt. The model supports object descriptions, attribute-specific questions, grasping-state queries, and multi-attribute questions.

To reduce the distribution gap between descriptive-text pretraining and question-based inference, QA-aware data augmentation is employed. For each sample, the augmentation produces 44 training variants: one original structured description without an accompanying question and 43 question/instruction–answer variants. These variants cover descriptive instructions, object identity, individual semantic attributes, grasping state, and multi-attribute queries. The exact templates used in the experiments are listed in Table S7.

A. QA-Aware Augmentation Templates

The 44 training variants used for each sample are listed below. Braced terms denote sample-specific annotation fields. The first variant corresponds to the original structured description and contains no question, whereas the remaining 43 variants form question/instruction–answer pairs.

The 44 variants comprise one answer-only original description and 43 question/instruction–answer variants. Specifically, they include four instruction-based descriptions, four object-identity questions, five material questions, four questions each for shape, size, hardness, color, and purpose, two grasping-state questions, and eight multi-attribute questions.

For quantitative evaluation, the model generates a structured description for each of the 15,085 Soma test samples. In addition to Exact Match, normalized Edit Distance, and BLEU-4 reported in the main paper, we evaluate field-level attribute accuracy. Generated descriptions that cannot be parsed into the complete structured template are counted as incorrect for the corresponding attribute fields.

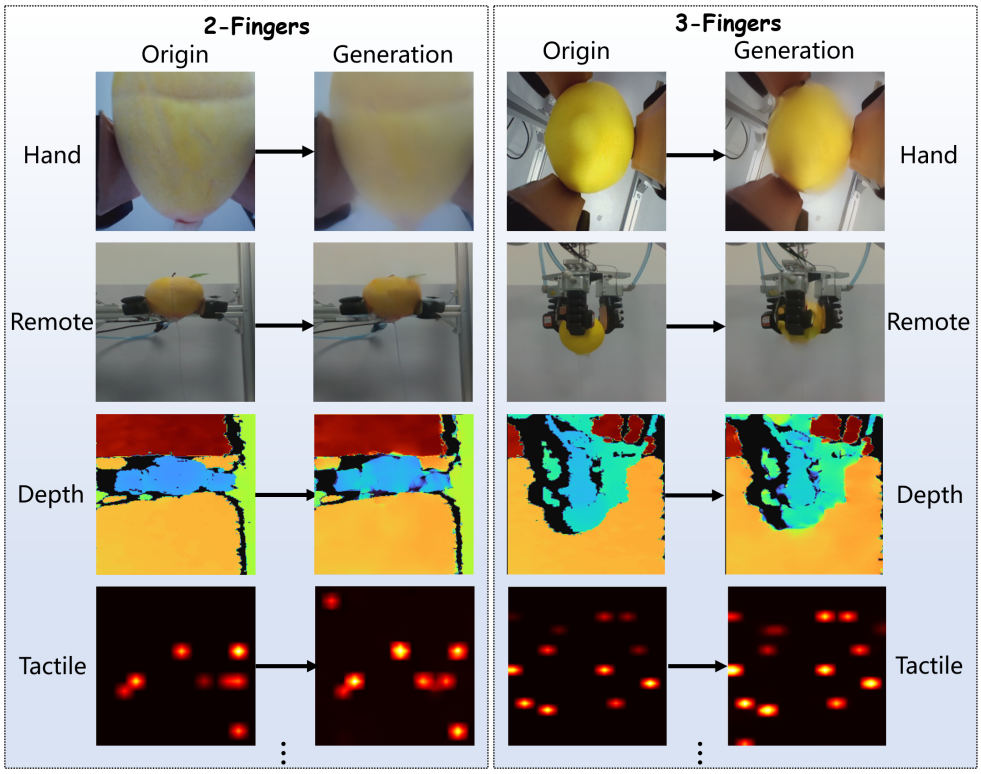


Fig. S2. Qualitative leave-one-out conditional generation results. Hand RGB, Remote RGB, Depth, and Tactile observations are generated using aligned features from the other five modalities. Examples cover two representative objects and both the two-finger and three-finger soft-gripper configurations.

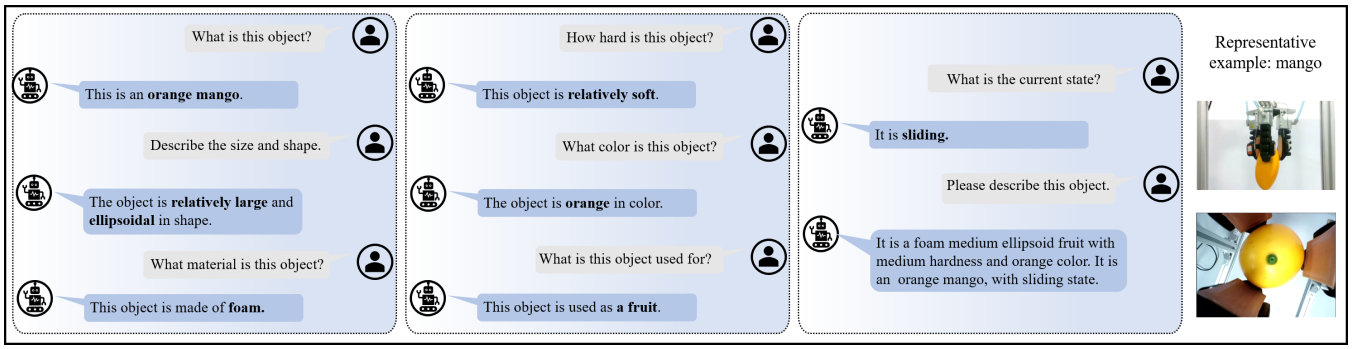


Fig. S3. Qualitative multimodal question-answering results for a representative mango sample using five sensory modalities. Examples cover object identity, physical attributes, functional purpose, grasping state, and free-form description. Key predicted attributes are highlighted in bold. The model may still exhibit limitations in response consistency.

As shown in Table S8, all seven fields achieve accuracies above 94%, with a pooled accuracy of 98.41% over all sample–attribute pairs. Field-level accuracy can exceed the Exact Match rate because Exact Match requires the entire structured description to match the reference, whereas field-level evaluation gives credit to correctly predicted attributes within otherwise nonidentical descriptions.

Across gripper configurations, the two-finger and three-finger subsets achieve Exact Match rates of 92.46% and 91.55%, respectively, with pooled attribute accuracies of 98.38% and 98.44%. The similar results indicate that the language-based evaluation remains consistent across the two soft-gripper configurations.

VIII. Additional Inference Complexity and Runtime Analysis

SomaAlign representation inference retains only the modality-specific encoders required for the six sensing modalities, while training-only auxiliary components are not required during deployment. The resulting inference model contains 1,332.2M active parameters. Table S9 summarizes the computational complexity, runtime, and parameter-storage requirements.

FP16 inference reduces the measured single-sample encoding latency from 97.98 to 43.67 ms, corresponding to approximately 22.9 samples per second. At batch size 32, FP16 encoding requires 745 ms per batch, equivalent to an amortized latency

TABLE S7

TRAINING VARIANTS USED FOR QA-AWARE AUGMENTATION. BRACED TERMS DENOTE SAMPLE-SPECIFIC FIELDS. ITEM 1 IS THE ORIGINAL STRUCTURED DESCRIPTION WITHOUT AN ACCOMPANYING QUESTION; ITEMS 2–44 ARE QUESTION/INSTRUCTION–ANSWER VARIANTS.

Index	Category	Question/instruction template	Expected answer field(s)
1	Free-form description	`[no question; original_desc only]`	All structured description fields
2	Instruction description	`Describe this object.`	Full original description
3	Instruction description	`What can you tell me about this object?`	Full original description
4	Instruction description	`Please describe this object in detail.`	Full original description
5	Instruction description	`Give me a full description of this object.`	Full original description
6	Material	`What material is this object?`	Material
7	Material	`What is this object made of?`	Material
8	Material	`Tell me the material of this object.`	Material
9	Material	`Describe the material of this object.`	Material
10	Material	`Is this object made of {material}?`	Material; affirmative answer
11	Shape	`What shape is this object?`	Shape
12	Shape	`What is the shape of this object?`	Shape
13	Shape	`Describe the shape.`	Shape
14	Shape	`Tell me the shape of this object.`	Shape
15	Size	`What is the size of this object?`	Size
16	Size	`How big is this object?`	Size
17	Size	`Describe the size.`	Size
18	Size	`Is this object large or small?`	Size
19	Hardness	`How hard is this object?`	Hardness
20	Hardness	`Describe the hardness.`	Hardness
21	Hardness	`Is this object hard or soft?`	Hardness
22	Hardness	`Tell me the hardness of this object.`	Hardness
23	Color	`What color is this object?`	Color
24	Color	`Describe the color.`	Color
25	Color	`What is the color of this object?`	Color
26	Color	`Tell me the color.`	Color
27	Purpose	`What is this object used for?`	Purpose
28	Purpose	`Describe the purpose of this object.`	Purpose
29	Purpose	`What is the purpose?`	Purpose
30	Purpose	`Tell me what this object is used for.`	Purpose
31	Object identity	`What is this object?`	Object identity
32	Object identity	`Identify this object.`	Object identity
33	Object identity	`Name this object.`	Object identity
34	Object identity	`What object is this?`	Object identity
35	Grasping state	`What is the current state of this object?`	State
36	Grasping state	`Is this object moving or stationary?`	State
37	Multi-attribute	`Describe the material and shape.`	Material, shape
38	Multi-attribute	`What are the material and color?`	Material, color
39	Multi-attribute	`Tell me about the shape and hardness.`	Shape, hardness
40	Multi-attribute	`Describe the material and size.`	Material, size
41	Multi-attribute	`What are the color and shape?`	Color, shape
42	Multi-attribute	`Tell me the size and shape.`	Size, shape
43	Multi-attribute	`Describe the material and hardness.`	Material, hardness
44	Multi-attribute	`What are the color and hardness?`	Color, hardness

of approximately 23.3 ms per sample. These measurements characterize representation encoding only and do not include downstream probe or task-specific model computation.

TABLE S8
FIELD-LEVEL ACCURACY OF MULTIMODAL QUESTION ANSWERING ON THE SOMA TEST SPLIT (%).

Attribute	**Accuracy**
Material	99.54
Size	99.16
Shape	98.93
Hardness	99.07
Color	98.67
Purpose	99.40
State	94.10
Pooled attribute accuracy	**98.41**

TABLE S9
COMPUTATIONAL COMPLEXITY AND RUNTIME OF SIX-MODALITY REPRESENTATION ENCODING WITH SOMAALIGN.

Metric	**Value**
Active inference parameters	1,332.2M
FLOPs per six-modality sample	1.666 TFLOPs
FP32 latency / sample	97.98 ms
FP16 latency / sample	43.67 ms
FP32 latency / batch-32	2498 ms
FP16 latency / batch-32	745 ms
FP32 parameter storage	5.33 GB
FP16 parameter storage	2.66 GB